MILITARY SPECIFICATION

HEAD ASSEMBLY, TANK PERISCOPE: 10513392

1. SCOPE

1.1 This specification covers one type of periscope head assembly which deflects a line of sight to the body assembly of a tank periscope. A rotatable mirror provides for elevating and depressing the field of view (SEE 6.1) .

2. APPLICABLE DOCUMENT

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830 — Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assemb}y, and Inspection of

MIL-13926 — Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

MIL-P-14232 — Park, Equipment and Tools for Ordnance Materiel; Packaging of

MIL-I-45607 — Inspection Equipment, SupplY and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 — Sampling PrOCedure and Tables for Inspection by Attributes

MIL-STD-109 — Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F8565556 — Pressure Tester — Two Station

F10518392— Head Assembly :10513392

F10513500 — Mount Assembly

PACKAGING DATA SHEET

MIL-P-14232/P10513392 — Packaging of Head Assembly: 10513392

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in Con nection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMEMENTS

3.1 ***Fabrication.*** The head assembly" shall be manufactured in accordance "with Drawing F10513392 and drawing pertaining thereto.

3.2 ***General specifications.*** The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL F-13926 specified below.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.3 Environment

3.3.1 ***Storage temperatures.*** The head assembly shall show no evidence of physical failure such as glass breakage and bonding

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or casting failure due to storage at temperatures ranging from -80° to +160° Fahrenheit (F). Subsequent to meeting this requirement, the head assembly shall be capable of meeting the requirements of 8.4.

3.3.2 ***Extreme operating temperatures.*** The head assembly shall meet the applicable requirements of 3.12 while exposed to and thermally stabilized at temperatures of 40° and +150°F.

3.3.3 ***Standard temperature.*** Unless otherwise specified herein, the head assembly shall meet the following requirements while exposed to a temperature between 60° and 90° F.

3.4 ***Vibration.*** The head assembly shall . show no evidence of physical failure due to being vibrated at 30 cycles per second and an amplitude of 0.06 inch, 0.12 inch total excursion (see 8.7.1). Subsequent to meeting this requirement, the head assembly shall be capable of meeting the following requirements.

3.5 ***Cleanliness.*** The head assembly shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than four particles of dirt visible. on the optical surfaces. The area of each particle shall not be in excess of an area equivalent to a size 10 dig. There shall be no foreign matter within the head assembly which interferes with observation or adversely affects image quality.

3.6 ***Sealing.*** The head assembly shall withstand an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) with no evidence of leakage.

3.7 ***Orientation.*** The requirements of 3.7.1 to 3.7.7 inclusive shaIl be met with the head assembly positioned as follows:

(a) Datum Plane "A", shown on DraWing F10513392, on a horizontal plane.

(b) Surface "C" of the input coupling, shown on Drawing - F10513392, positioned 66° with respect to Datum plane "A".

3.7.1 ***Collimation.***

3.7.1.1 **Pre-vibration.** An incident line of sight which is originally perpendicular to Datum Plane "B", shown on Drawing F10513392, shall emerge perpendicular to Datum Plane "A" within 0.20 mil.

3.7.1.2 ***Post-vibration.*** Subsequent to vibration, an incident line of sight which is originally perpendicular to Datum Plane "B" shall emerge perpendicular to Datum Plane "A" within 0.20 mil and remain within 0.10 mi1 of the collimation reading established in 3.7.1.1.

3.7.2 ***Collimation change (due to assembling to mount).*** An original line of sight shall not deviate more than 0.10 mil due to repeated. assembling of the head assembly to the mount with which the head assembly is intended. to be used.

3.7.3 ***Image tilt.*** The image of a target line originally perpendicular to Datum Plane "A" shall emerge perpendicular to Datum Plane "B" within 30 minutes of arc. .

3.7.4 ***Excursion range.*** Rotation of the input coupling shall cause the line of sight to deflect a minimum of 22.0° above to 18.0° below the horizontal plane.

Note. At no time shall the input coupling be allowed to strike hard against the travel stops.

3.7.5 ***Plumb travel.*** Starting with the line of sight in coincidence with a vertical target line at the horizontal position, the lime of sight shall track the vertical target line within 0.25 mil when elevated and depressed through the range specified in 8.7.4.

3.7.6 ***Elevation accuracy.*** This requirement shall be met with backlash removed and the angle measured in a vertical plane. The angle formed by the line of sight and Surface "C"

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of the input coupling shall remain constant within 0.20 mil at any elevation setting from 18° depression to 22° elevation.

3.7.7 ***Backlash.*** Backlash between the line of sight and the input coupling shall not exceed 0.05 mil.

3.8 ***Resolution.*** Resolution shall be 6.0 seconds of arc or less when scanning the exit aperture with 24-power telescope having a 3.0 inch entrance pupiL

3.9 ***Spherical power.*** Spherical power shall not exceed 2.0 milidiopters when scanning the exit aperture with a 24-power telescope having a 3.0 inch entrance pupil.

3.10 ***Astigmatism.*** Astigmatism shall not exceed 1.0 millidiopter when scanning the exit aperture with a 24-power telescope having a 3.0 inch entrance pupil.

3.11 ***Coupling travel.*** Rotation of the input oupling shall be limited by contact with the wavel stop pin only.

3.12 ***Torque.*** The running torque required to rotate the input coupling through the range specified in 3.7.4 shall be within the limits of 3.0 and 7.0 inch-pounds at standard temperature. When checked at the temperatures specified in 3.3.2, the running torque shall not exceed 11.0 inch-pounds.

3.13 ***Interior atmosphere*** The interior atmosphere of the head assembly shall contain nitrogen having a dewpoint at least as low as –25° F.

4. QUALITY ASSURANCE PROVISIONS

4.1 ***Responsibility for inspection.*** Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 ***General provisions.*** The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. inspection of components and subassemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 ***Inspection provisions.***

4.2.1 ***Submission of product.*** Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MXL-STD-105.

4.2.2 ***Examination and tests.*** Components and subassemblies shall be inspected in ac-- cordance with the inspection provisions con* tained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAp) listed in the List of Inspection Documents, IEL 10513392. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with STD-105, and the inspection level and sampling plans specified in Table 1 titled "Classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

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TAble I. Classification of Defects

Use Inspection Level II of Table I with Sampling Plan Table II-A of MIL-STD-105

		Requirement	Test procedures
CRITICAL: None			
MAJOR: AQL 0.65% Defective			
101.	Collimation, pre-vibration	3.7.1.1	4.5.3
102.	Vibration	3.4	4.5.4
103.	Cleanliness	3.5	4.5.5
104.	Collimation, Post-vibration	3.7.1.2	4.5.8.1
105.	Collimatoin change (due to attaching body assembly)	3.7.2	4.5.8.2
106.	Image tilt	3.7.3	4.5.8.3
107.	Elevation accuracy	3.7.6	4.5.8.4
108.	Excursion range	3.7.4	4.5.8.5
109.	Plumb travel	3.7.5	4.5.8.6
110.	Backlash	3.7.7	4.5.8.7
111.	Resolution	3.8	4.5.9
112.	Spherical power	3.9	4.5.10
113.	Astigmatism	3.10	4.5.11
114.	Coupling travel	3.11	4.5.12
115.	Torque	3.12	4.5.13
116.	Sealing (+ 60° to 90° F)	3.6	4.5.6
117.	Interior atmosphere (+ 60° to 90° F)	3.13	4.5.7
MINOR: None			

Note. Tbe tests of Table I shall be conducted at a tempera-are between plus 60° and 90° F.

4.2.3 ***Acceptance and rejection.*** Rejected lots shall be screened for all defective characteristics. Removal or correction of defected units and resubmittance of rejected lots shall be in accordance with "Acceptance and action" as specified in MIL-Std-105,

4.3 ***Control tests.***

4.3.1 ***General.*** One head assembly shall be elected at random as a control sample from ach 100 produced. The control sample shall have successful] y met all other acceptance ests specified herein, and shall then meet the following requirements and tests:

Control test	Require-ments	Test procedures
01. Fabrication	8.1	visual, per drawings
02. General specification	3.2	Visual MIL-F-13926

4.3.2 ***Environmental.*** Three head assemblies shall be selected at random as control samples from each 50 produced, or from each 'months production, whichever occurs first. The control samples shall have met the requirements and tests of table I and shall then meet the following requirement and tests:

Control test	Require-ments	Test procedures
808. Storage temperatures (-80 & +160°F)	3.3.1	4.6.1
804. Extreme operating temperatures (40° & +150°F)	3.3.2	4.6.2

4.3.3 ***Control test failure.*** Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.8.2).

4.4 ***Inspection equipment.*** Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 ***Government furnished inspection equipment,*** Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 ***Contractor furnished inpectiion equipment.***

4.4.2.1 ***Government design.*** All inspection equipment required by drawings forming a part of the contract and not provided by the Govrnment shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents IEL 10513892.

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4.4.2.2 ***Contractor design.*** The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit and will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.3 ***Inspection equipment design requirements.***

4.4.2.3.1 ***Sealing.*** Special pressure testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester – Two Station.

4.4.2.3.2 ***Special testing equipment utilized for orientation of head assembly.*** The testing equipment utilized to mount and orient the head assembly for the accuracy tests shall permit mounting Datum Plane "A", as shown on Drawing FI0513392, in a horizontal plane within 10 seconds, (0°0′10″) and Surface "C" of the input coupling, as shown on Drawing F10513392, positioned 66° within 10 seconds (0°0′10″) with respect to Datum Plane "A". The testing equipment shall also provide for articulation of Surface "C" of the input coupling to any angular position between its travel stops. The necessary locating surfaces and head mounting features of the mount Assembly Drawing F1051350O, shall be incorporated into the testing equipment.

4.4.2.3.3 ***Special testing equipment to test collimation, plumb travel, elevation accurracy, and backlash.*** The testing equipment for these tests shall be accurate within 5 seconds (0°0'5").

4.4.2.3.4 ***Sspecial testing equipment to test image tilt.*** The testing equipment for this test shall be accurate within 3 minutes.

4.4.2.3.5 ***Special testing equipment to test excursion range.*** The testing equipment utilized for these tests shall be calibrated to the limits of 22° + 1° above and 18° + 1° below the original line of sight projecting from the head assembly.

4.4.2.3.6 Rigidity, stability, ease of maintenance, and adaptability of setup and calibration equipment with complete provision are also prime requirements. All setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become part of the testing equipment.

4.5 ***Test methods and procedures.***

Note. Caution shall be exercised during the following tests to prevent the input coupling from striking hard against the travel stops.

4.5.1 ***Storage temperatures.*** Test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The head assembly shall be. exposed to each specified temperature extreme for a minimum of 5 hours. After the temperature exposures, the head assembly shall be stabilized in an ambient temperature between 60° and 90° Fahrenheit and subjected to the visual, tactile, and audible examination of 4.5.4. When the head assembly has passed the above examinations, it shall then be subjected to the tests specified in 4.5.3 and 4.5.4 to determine overall compliance with 3.3.1. Caution shall be exercised during this test to avoid subjetting the head assembly to thermal shock.

4.5.2 ***Extreme operating temperatures.*** Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.2. The head assembly shall be exposed to each temperature extreme for a period of 5 hours. At the end of each 5 hour period and

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while stiIl at the temperature extreme the head assembly shall be subjected to the test specified in 4.5.13 and shall conform to the applicable requirements of 3.3.2 for each temperature extreme. Caution shall be exercised during this test to avoid subjecting the head assembly to thermal shock.

4.5.3 ***Collinnation, pre-vibration.*** Prior to the vibration test, position the head assembly as specified in 4.5.8 and position Surface "C" of the input coupling 66° with respect to Datum Plane "A". Direct a line of sight through the viewing collimator to the target. The intersecting point of the vertical and horizontal reticle lines of the viewing collimator shall be superimposed on the intersecting point of the vertical and horizontal reticle lines of the target within the tolerance specified in 3.7.1.1, as measured on the viewing collimator. Record this measurement for use in the subsequent test in 4.6.8.1. The head assembly shall then be subjected to the test Specified in 4.5.4

4.6.4 ***Vibration.*** Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.4 should be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. Immobilize the input coupling to any position that shall prevent it from contacting the travel stops. The head assemby shall be vibrated in a verticle plane for a period of 6 minutes ± 16 seconds. Subsequent to vibration, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 8.4 prior to being subjected to the remaining tests specified herein. A visual examination of the head assembly shall be made for missing or damaged parts, chipped or loose paint, and corrosion. Minimum visual examination shall include the windows, articulated head mirror, input coupling, mounting surfaces, screws and sealant. Optical characteristics Shall be examined by viewing through the entrance and exit windows of the head assembly to detect excessive dirt, scratches, digs, fractures or foreign matter which would interfere with the optical system, and be cause for rejection. A tactile examination shall also be conducted to check functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall be made by positioning the ear close to the head assembly and while vigorously shaking the head assembly with both hands listen for loose internal optical and mechanical to parts (mirror, windows, screws, etc). Anyone of the possible defects defined above shall be cause to consider a head assembly defective, unless separate defects – per hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Head assemblies which have passed the above examinations shall be subjected to the tests of 4.6.6 to 4.5.13 inclusive to determine overall compliance with 8.4.

4.6.6 ***Cleanliness.*** The optical surfaces of the head assembly shall be inspected for Cleanliness in accordance with MIL-O-13830. The head assembly shall conform to the cleanliness requirements of 3.5.

4.5.6 ***Sealing.*** The internal Pressurizing of the head assembly, to test seating, shall be performed with special testing equipment conforming to 4.4.2.8.1. The pressure shall be applied to the head assembly through the valve provided- in the upper housing. Pressurize the head assembly with dry nitrogen gas, having a dewpoint at least as low as minus 25° Fahrenheit, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the head assembly for a period of 5 minutes ± 15 seconds. During this time period; the input coupling shall be rotated through one full excursion of its articulation range. At the end of-the 5-minute period there shall be no evidence of leakage, as indicated on the pressure gage of the testing equipment to determine compliance with 3.5.

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4.5.7 ***Interior atmosphere.*** A dewpoint tester shall be used to test the dewpoint of the inkier atmosphere of the head assembly. Upon satisfactory completion of the sealing test the head assembly shall be bled to 1 psig and and flushed with dry nitrogen gas through a regulator that will prevent the internal pressure, in the head assembly, from exceeding the 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the head assembly has a dewpoint at least as low as minus 25° Fahrenheit, as measured with the dewpoint tester. The head assembly shall then be sealed. The internal atmosphere of the head assembly shall conform to the requirements of 3.13.

4.5.8 ***Orientation.*** The following tests shall be performed with the head assembly positioned on special testing equipment that conforms to the requirements of 4.4.2.3.2. The testing equipment shall also provide a target projector collimator, a calibrated penta prism and a viewing collimator positioned as shown in fig. 1. The collimators shall have an entrance pupil of at least 2.0 inches and shall be aligned to the central aperture (geometric axis) of the head assembly. The viewing collimator positioned at the penta prism shall be adjusted for parallel light and shall be auto-collimated to Datum Plane "A" as shown on Drawing F10513392. The viewing collimator shall be capable of measuring the required optical accuracies of the head assembly. Adjust the target projector collimator for parallel light. Auto-collimate this target projector collimator to Datum Plane "B", and align its vertical and horizontal reticle lines coincident to the vertical and horizontal reticle lines of the viewing collimator. The target projector collimator shall have an appropriate calibrated target reticle pattern for projection into the optical system of the head assembly. The vertical reticle lines of the collimators shall be adjusted plumb. Auto-collimation and alignment shalI be accomplished with the head assembly removed from the line of sight through the collimators.

4.5.3.1 ***Collimation post-vibration.*** This test shall be performed subsequent to vibration. Postion Surface "C" of the input coupling 66° with respect to Datum Plane "A". Perform this test in accordance with 4.5.3. Record teh collimation measurement. Any difference from the recorded measurement established in 4.5.3 shall conform to the requirements of 3.7.1.2 as determined by the recorded values. The head assembly shall then be subjected to the test in 4.5.8.2.

4.5.8.2 ***Collimation change (due to assembling mount).*** Subsequent to the collimation, post-vibration test the head assembly shall be removed from the testing equipment and repositioned as in 4.5.8.1. Repeated removal and positioning of the head assembly, a minimum of three times, shall be performed. After each positioning the head assembly shall be inspected in accordance with 4.5.3 and each collimation measurement recorded. The head assembly shall conform to the requirements of 3.7.2 as determined by the recorded values at each positioning.

4.5.8.3 ***Image tilt.*** Position surface "C" of the input coupling 66° with respect to Datum PIane "A". Direct a line of sight through the viewing collimator to the target image. The image of the vertical target line shall be paraIel to the verticle reticle line of the viewing collimator within the tolerance specified in 3.7.3 as measured on the target.

4.5.8.4 ***Elevation accuracy.*** The testing equipment specified in 4.5.8 shall be utilized for this test except, the target projector collimator is not required. Auxiliary testing equipment employing an auto-collimation device, affixed to the input coupling of the head assembly shall be used in place of the target projector collimator. This testing equipment shall consist of an arm assembly with a calibrated optically fiat front surface mirror and an appropriate adapter that accommodates the input coupling of the head assembly positioned as shown in fig. 2. The minor shall be auto-collimated to the viewing collimator when Surface "C" of the input coupling, of

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the head assembly, is positioned 66° with respect to Datum Plane "A". The reflecting (auto-collimating) surface of the mirror shall lay in a plane parallel to Datum Plane "B" and a vertical center line passing the geometric axis of the shaft of the input coupling adapter. The reflecting (auto-collimating) surface of the mirror shall also remain parallel to the vertical center line of the input coupling adapter during rotation through its full excursion of travel. Auto-collimation of the mirror and the viewing collimator shall be accomplished with the head assembly removed from the line of sight. Position and secure the head assembly on the holding device with the auxiliary testing device affixed and orientated on the input coupling of the head assembly. With backlash removed for elevation travel, position Surface "C': of the input coupling, with respect to Datum Plane "A", so that the line of sight is deflected 18° below the horizontal plane. Direct a line of sight through the viewing collimator and note the position of the intersecting point of the viewing collimator's reticle lines. While observing the boresight point (intersecting point of the vertical and horizontal reticle lines) of the real and reflected image rotate Surface "C" of the input coupling, with respect to Datum Plane "A", so that the line of sight is deflected from 18° below the horizontal plane to 22° above the horizontal plane. During this rotation of the input coupling the reflected reticle image shall not deviate from the real reticle image beyond the tolerance specified in 3.7.6, as measured on the viewing collimator.

4.5.8.5 ***Excursion range.*** Perform this test in conjunction with the test in 4.5.8.4 (elevation accuracy). Rotation of the input coupling shall conform to the requirements specified in 3.7.4.

4.5.8.6 ***Plumb travel.*** Perform this teat in sequence to the test in 4.5.8.4. Position Surface "C" of the input coupling at 66° with respect to Datum Plane "A". While observing the boresight point of the real and reflected image rotate Surface "C" of the input coupling the maximum elevation, then to maximum depression. During this rotation, the reflected image shall not deviate in azimuth from the real. image beyond the. tolerance. specified in 3.7.5.

4.5.8.7 ***Backlash.*** Perform this test in sequence with 4.5.8.6. With backlash removed for direction of travel, position Surface "C" of the input coupling at 66° with respect to Datum Plane "A". Record the position of the boresight point of -the reflected image, as measured on the viewing colliniator. Rotate Surface "C". Of the input coupling approximately 10° in the same direction of travel, then reverse rotation of the input coupling (with backlash included) until Surface "C" of the input coupling is repositioned 66° with respect to Datum Plane "A"," without overtravel. Record this position of the boresight point of the reflected image, as measured on the viewing collimator. Any difference between the recorded readings is the amout of backlash. Backlash shall not exceed the limit specified in 8.7.7.

Note. During this operation (4.5.8.7), if overtravel occurs. the test shall be nullified and the procedure specified for backlash shall again be pew formed.

4.5.9 ***Resolution.*** Position the head assembly on its side, with surface "C" of the input coupling immobilized at 66° with respect to Datum Plane "A". The test for resolution shall be performed utilizing an auxiliary telescope conf orming to the requirements of 3.8, a resolving power chart on a wall target representing the angular subtense for the seconds of arc specified for resolution, and a supporting medium to receive this test equip ment as well as the item to be tested. The test method and procedure shall be in compliance with the "Resolution Test" specified in MIL-O-13830, and the resolving power chart shall contain four line sets as shown in fig. 1 therein. Resolution shall be within the value specified in 3.8, as determined by viewing through the auxilary telescope while scanning the longitudinal axis of the clear exit aperture of the Window, 8215997, of the head assembly.

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4.5.10 **Spherical power.** With the head assembly positioned as in 4.6.9 and utilizing the resolving power chart specified therein, the test for spherical power shall be performed with the aid of a calibrated milli-dioptometer conforming to the requirements of 3.9. Set the millidioptometer scale to zero milli-diopters and adjust the milli-dioptometer eyepiece for best focus of its reticle. Position the milli-dioptometer so that its optical axis is perpendicular to the exit aperture of the Window, 8215997, of the head assembly. Adjust the milli-diopter scale for best focus of the numeral in the center of the resolving power chart. Note the milli-diopter value. Scan the longitudinal axis of the clear exit aperture of Window, 8215997. Best focus of the numeral at any point in the scanning area shall not exceed the tolerance (milli-diopters) specified in 3.9, as measured on the milli-diopter scale.

4.5.11 ***Astigmatism.*** Perform this test in accordance with 4.5.10, except, adjust the milli-diopter scale for best focus of each of the four line sets. Any difference between the milli-diopter values, obtained as a result of best focus of each of the four line sets, is the amount of astigmatism. Astigmatism at any point in the scanning area shall not exceed the tolerance (rnilli-diopters) specified in 3.10, as measured on the milli-diopter scale.

4.5.12 ***COupling travel.*** Rotation of the input coupling through its full excursion of travel shall be subjected to a tactile examination while observing the input coupling and the travel stop. Coupling travel shall conform to the requirements of 3.11.

4.5.13 ***Torque.*** Standard torque measuring equipment capable of measuring the torque values specified in 3.12 shall be utilized for this test. An appropriate adapter, to accommodate the input coupling shall be used with the torque measuring equipment. Torque shall be measured while rotating the input coupling through its full range of travel. The running torque required to rotate the input coupling shall conform to the applicable requirements of 3.12 for the temperatures indicated.

5. PREPARATION FOR DELIVERY

5.1 ***Packaging, packing and marking.*** Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14282/ P10513392. The level of protection shall be as specified in the procurement document.

6, NOTES

6.1 ***Intended use.*** The Tank Periscope Head Assembly 10513392 is primarily intended for use as the head assembly for the gunner's periscope in the XM551 Vehicle. Assembled to the Tank Periscope Body Assembly, the head assembly is a component of the unity-power daylight, nine-power passive XM44 Tank Periscope.

6.2 ***Ordering data.*** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and pack. ing.

(c) Applicable stock number.

(d) Applicable packaging data sheet number (see 5.1).

Custodian:
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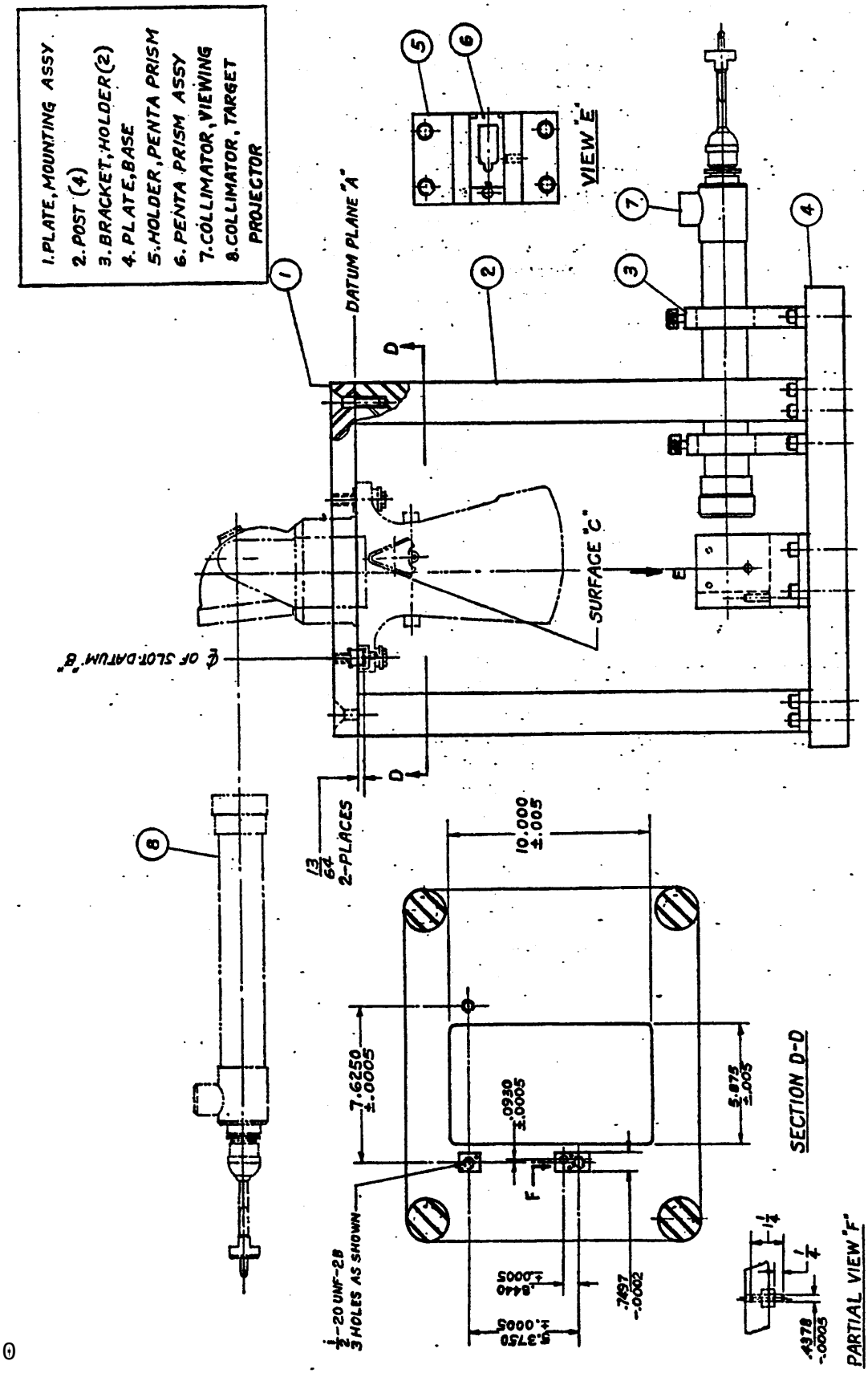


FIGURE 1. Accuracy test

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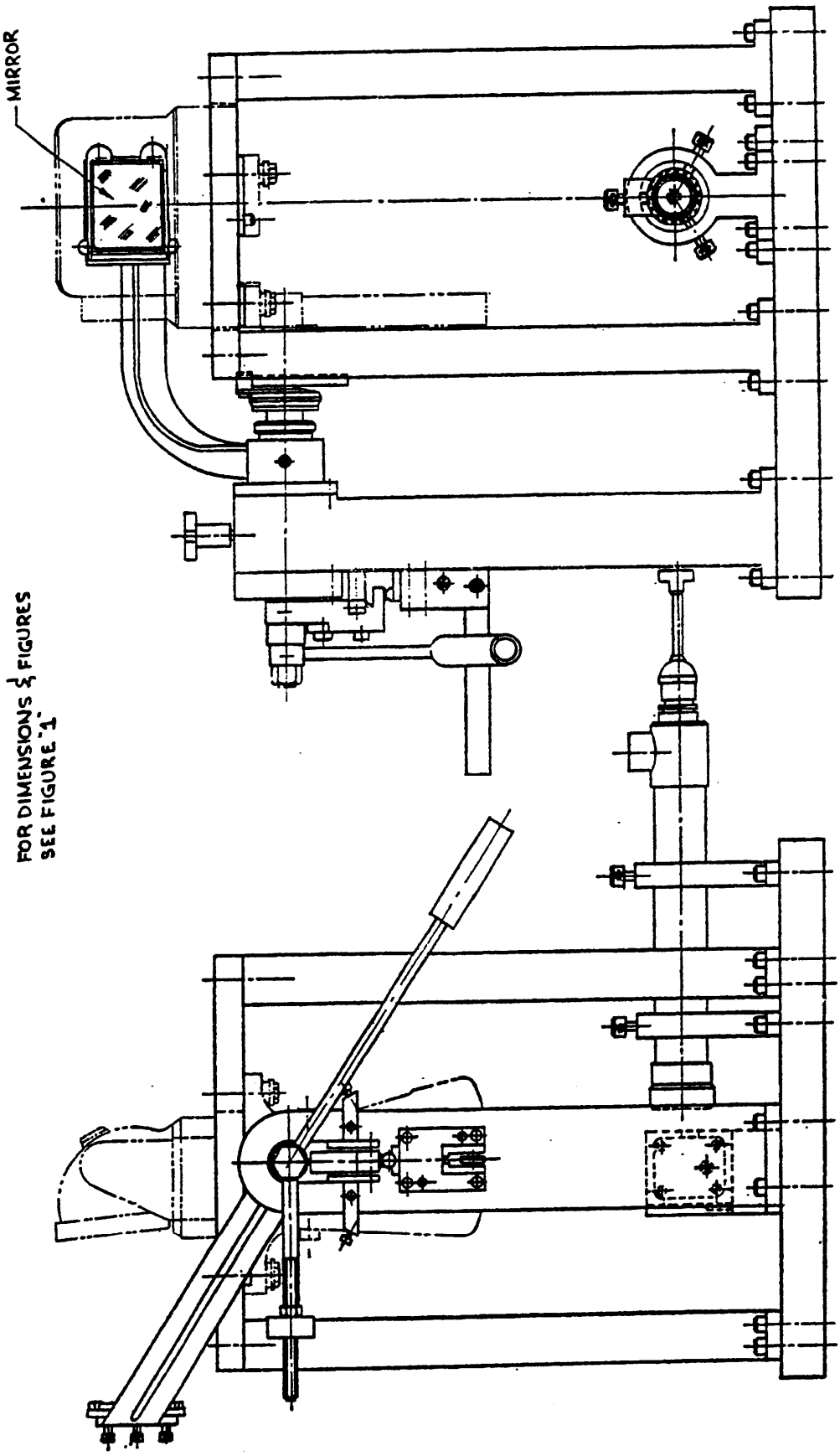


FIGURE 2. *Accuracy test*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity).

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 61 1426

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Philadelphia, Pa. 19137

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-5500
Philadelphia, Pa. 19137

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MILITARY SPECIFICATION

HEAD ASSEMBLY, TANK PERISCOPE: 10513392

This amendment forms a part of Military Specification MIL-H-60153(MU), dated 10 February 1965.

Paragraph 3.7.1.1: Delete paragraph and substitute:

* "3.7.1.1 Pre-vibration.- This requirement shall be met prior to meeting the requirements of 3.4. An incident line of sight which is originally parallel to Datum Plane "A" and perpendicular to the intersection of Datum Planes "A" and "B", shown on Drawing F10513392, shall emerge parallel to Datum Plane "B" within 0.20 mil and perpendicular to the intersection of Datum Planes "A" and "B" within 0.60 mil."

Paragraph 3.7.1.2: Delete paragraph and substitute:

* "3.7.1.2 Post-vibration.- This requiraement shall be met subsequent to meeting the requirements of 3.4. The incident line of sight specified in 3.7.1.1 shall emerge within 0.10 mil of the collimation readings established in 3.7.1.1."

Paragraph 3.8: Delete paragraph and substitute:

"3.8 Resolution. - Resolution shall be 2.0 seconds of arc or less when observing with a telescope of at least 90 power. The entrance pupil of the telescope shall be not less than 7.0 inches. Resolution shall be observed at one focus setting of the telescope."

Paragraphs 3.9 and 3.10: Delete paragraphs and substitute: .

* "3.9 Radiant transmittance. - Minimum luminous transmittance and minimum transmittance at 0.85 micron shall be not less than 90 percent."

Paragraph 4.2.2, Table I: Delete characteristics 112 and 113. .

Paragraph 4.3.2: Delete paragraph and substitute the following:

"4.3.2 Environmental. - Three head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. The samples shall not be selected from those which have previously been vibrated in Table I. They shall be subjected to the following sequence of tests to determine compliance with their respective requirements:

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CONTROL TESTS	REQUIREMENTS	TEST PROCEDURES
303. Storage temperatures (-80° and +160°F)	3.3.1	4.5.1 ,
304. Operating temperatures (-40° and +150°F)	3.3.2	4.5.2
305. Vibration 1/ (60° to 90°F)	3.4	405.4"

1/ Subsequent to being vibrated the head assembly shall be subjected to the tests of 4.5.5 to 4.5.13 inclusive to determine overall compliance with 3.4.

* Paragraph 4.5.8, 4th sentence: Delete and substitute:

"The viewing collimator positioned at the penta prism shall be adjusted for infinity focus and auto-collimated to Datum Plane "A", in elevation, and the intersection of Datum Planes "A" and "B" in azimuth."

* Paragraph 4.5.8, 6th sentence: Delete and substitute:

"The objective lens of the target projector collimator shall be focused for infinity."

* Paragraph 4.5.8, 7th sentence: Delete and substitute: .

"Establish the optical axis of the target projector collimator parallel to Datum Plane "A" and perpendicular to the intersection of Datum Planes "A" and "B" by means of auto-collimation. Align the vertical and horizontal reticle lines of the target projector collimator to the vertical and horizontal reticle lines of the viewing collimator."

* Paragraph 4,5.8.4, 5th sentence: After "passing" insert "through".

Paragraph 4.5.9, last sentence: Delete and substitute: .

"Resolution shall be within the value specified in 3.8 as determined by observing through the exit aperture of Window 8215997 with auxiliary telescope."

Paragraphs 4.3.10 and 4.5.11: Delete paragraphs and substitute:

* "4.5.10 Radiant transmittance.

4.5.10.1 Luminous transmittance.- Perform this test with the head assembly positioned as in 4.5.9. A diffused luminous source of a color temperature of 2870°K shall be observed with an image forming photometer. The transmittance measurement shall be made

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while scanning the longitudinal axis of the clear exit window aperture. The transmittance in the scanning area shall meet the value specified in 3.9, measured on the photometer. Compute the percentage of transmittance utilizing the following formula:

$$\frac{\text{Luminance of source}}{\text{Luminance of source as seen through head}} \text{ X100}$$

4.5.10.2 Transmittance, 0.85 micron.- Conduct this test in sequence to the test in 4.5.10.1. A diffused light source of approximately 100 foot-lamberts luminance shall be utilized on a target at a distance of 10 feet from the head assembly. The target shall be observed while scanning the longitudinal axis of the clear exit window aperture with the aid of an image forming radiometer. The radiometer shall be filtered for 0.85 micron radiation by a filter which has a peak transmission at 0.85 micron and a bandwidth of ±.01 micron at 1/3 of peak. The transmittance in the scanning area shall meet the value specified in 3.9, as measured on the radiometer. Compute the percentage of transmittance utilizing the following formula:

$$\frac{\text{Filter radiometer reading of target}}{\text{Filter radiometer reading as seen through head"}} \text{ X100\%}$$

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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Preparing activity:

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